

Paper RQM- $\hbar$ — $\hbar$ -Origin in Chain-Step Participation Quantum

A Generative Paper of the Event Density Program

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of $\hbar$ from nothing, nor a first-principles numerical-value derivation. Result is conditional on the 13 ED primitives (Paper_087) + declared paper-specific postulates.
2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set), not the operational-reconstruction lineage.
3. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is stated in §1.
4. It does **not** derive the numerical value $\hbar \approx 1.055 \times 10^{-34}$ J·s from first principles. The substrate bandwidth-density $\mathcal{B}_{P04}$ has no independent substrate definition in the current corpus; per the §3.3 circularity disclaimer, $\hbar = \int \mathcal{B}_{P04} ds$ is a substrate-side relabeling of $\hbar$, not an independent computation.
5. The **dimensional structure** of $\hbar$ as action = bandwidth $\times$ time and its identification as a substrate-invariant participation quantum per chain-step are FORM-FORCED by P04 + P05 + P11 + P03 + P13 homogeneity; the **numerical value** of $\hbar$ (and of $\mathcal{B}_{P04}$, chain-step duration τ) is INHERITED via empirical matching.

Abstract

This paper identifies the substrate-level structural origin of Planck's constant $\hbar$ as the **participation quantum per chain-step**: the P04 bandwidth content integrated over a single substrate edge-traversal under P05 polarity-transport. The dimensional structure (action = energy $\times$ time = bandwidth $\times$ time) is FORM-FORCED by the substrate primitives, and chain-step invariance under P03 + P13 homogeneity forces $\hbar$ to be a universal constant. The numerical value is INHERITED via empirical matching; an independent substrate definition of the bandwidth-density $\mathcal{B}_{P04}$ is OPEN (§3.3 circularity disclaimer). Verdict per Paper_095: **M3** (value-INHERITED with explicit structural-origin). Key falsifier line: **F4** (substrate derivation of $\hbar$'s value from first principles would upgrade the verdict from value-INHERITED to value-FORCED).

1 Statement of Result

Claim. Planck's constant $\hbar$ has substrate-level **structural origin** as the **participation quantum per chain-step**: the substrate-bandwidth content per unit increment of chain-edge traversal under P05 polarity-transport. Specifically:

$$\hbar = (\text{participation budget per chain-step}) = \int_{\text{step}} \mathcal{B}_{P04}(s) ds,$$

where $\mathcal{B}_{P04}(s)$ is the P04 bandwidth-density per substrate edge-element ds , and the integral is over a single substrate chain-step.

The **dimensional structure** of $\hbar$ (action = energy $\times$ time = bandwidth $\times$ time) is FORM-FORCED by P04 + P05 + P11. The **numerical value** $\hbar \approx 1.055 \times 10^{-34} \text{ J} \cdot \text{s}$ is INHERITED via empirical matching.

Verdict: **M3**. Structural form forced; value inherited.

2 Primitive Inputs

2.1 Primitives invoked

- **P01 (Chains)** — supplies chain structure on which chain-steps are defined.
- **P04 (Bandwidth)** — non-negative additive scalar; supplies the budget content per substrate cell.
- **P05 (Polarity-transport)** — supplies the edge-traversal under which chain-step is realized.
- **P11 (Commitment-irreversibility)** — supplies directional structure of chain-step.

2.2 Upstream Dependencies

- **I-002:** Born-Gleason rule — $\hbar$ -dependent probability normalization.
- **I-RQM-T8:** Canonical commutation relations — $\hbar$ as rate constant.
- **I-012:** P-RB-1 local rate of becoming — $\hbar$ in $E/\hbar$ atomic-superposition rate.
- **I-093:** T18 kernel-level arrow of time.
- **I-Action:** Standard action / phase-space machinery.

3 Derivation

3.1 Action dimensions and chain-step structure

Standard QM places $\hbar$ as the conversion factor between action (units: energy $\times$ time) and phase ($e^{iS/\hbar}$). Action has dimensions $[S] = \text{kg} \cdot \text{m}^2/\text{s} = \text{J} \cdot \text{s}$.

The substrate-level question: what is the substrate-level origin of this dimensional combination?

P05 (polarity-transport) describes edge-traversal: a substrate chain traverses edges with polarity rotating under P09 ($U(1)$ -valued polarity). Each chain-step (single edge traversal) accumulates:

- A finite increment of substrate time (per P11 commitment-irreversibility).
- A finite increment of polarity rotation (per P09 + P05).
- A finite participation budget consumption (per P04 bandwidth).

The **participation budget per chain-step** has dimensions of **bandwidth $\times$ time = action**. This is the substrate origin of $\hbar$'s dimensional structure.

3.2 Chain-step invariance

By P03 (channel + locus indexing, spatial homogeneity) and P13 (time homogeneity), the substrate is **homogeneous**: chain-steps at different loci and different times consume the same participation budget. The participation quantum per chain-step is a **substrate-invariant**.

This invariance is the substrate analog of the empirical observation that $\hbar$ is the same in every QM experiment, every laboratory, every epoch. The substrate-level statement: chain-step invariance under P03 + P13 forces $\hbar$ to be a universal constant.

3.3 $\hbar$ as the invariant quantum

Definition:

$$\hbar \equiv \int_{\text{one chain-step}} \mathcal{B}_{\text{P04}}(s) ds.$$

The integral is over a single substrate chain-edge traversal. $\mathcal{B}_{\text{P04}}(s)$ is the bandwidth density (P04) along the edge. By P03 + P13 invariance, this integral is the same across all substrate chain-steps; hence $\hbar$ is a universal constant.

By dimensional analysis:

- $[\mathcal{B}_{\text{P04}}] = \text{bandwidth} = \text{energy} / \text{time} = \text{J/s}$.
- $[ds] = \text{time} \times c$ (for null-edge traversal) or time (for time-like edge traversal); in either case, dimensions of time.
- $[\hbar] = (\text{J/s}) \times \text{s} = \text{J} \cdot \text{s}$.

The structural form of $\hbar$ as bandwidth $\times$ time is FORM-FORCED.

Circularity disclaimer. $\mathcal{B}_{\text{P04}}$ has no independent substrate definition in the current corpus; the equation $\hbar = \int \mathcal{B}_{\text{P04}} ds$ is a substrate-side **relabeling** of $\hbar$, not an independent computation. $\hbar$'s structural role (action-quantum coupling P04 $\leftrightarrow$ P13) is identifiable; $\hbar$'s numerical value is empirically INHERITED. An independent substrate definition of $\mathcal{B}_{\text{P04}}$ is OPEN.

3.4 Inheritance of the numerical value

The numerical value $\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s}$ is determined by:

- The substrate-bandwidth-density $\mathcal{B}_{\text{P04}}$ (substrate parameter; not derivable from primitives in this paper).
- The substrate chain-step duration (related to substrate scale ℓ_{ED} via c ; not derivable here).

The product of these substrate parameters gives $\hbar$. Both are INHERITED via empirical matching; the substrate framework does not derive $\hbar$ from first principles; it identifies the substrate-level structural origin.

This is the **H1-dominant value-INHERITED** regime for the fundamental constants (analogous to Arc-M H1-dominant for mass parameters).

3.5 Connection to chain-step participation in standard QM

The substrate-level identification of $\hbar$ with chain-step participation connects to several standard-QM facts:

De Broglie wavelength $\lambda = h/p$: a particle's wavelength equals h divided by momentum. Substrate-level: the chain-step participation budget per unit momentum carry; particles with larger momentum traverse more chain-steps per unit substrate length, giving shorter wavelength.

Energy quantization $E = n\hbar\omega$: harmonic-oscillator levels are equispaced by $\hbar\omega$. Substrate-level: each quantum represents one additional participation-budget-per-chain-step of oscillator excitation.

Bohr-frequency rule $\nu = (E_2 - E_1)/h$: substrate-level rate of becoming (Paper_012 P-RB-1) for atomic superpositions, where $E/\hbar$ is the rate of phase accumulation per substrate time.

In each case, $\hbar$ appears as the dimensional bridge between substrate-level chain-step structure and the macroscopic energy-time observables.

3.6 Why $\hbar$ is small

The substrate scale $\ell_{\text{ED}} \sim \ell_P$ is extremely small. The chain-step duration is $\tau \sim \ell_P/c \sim 10^{-43}$ s. The bandwidth-density $\mathcal{B}_{\text{P04}}$ is correspondingly large per substrate cell to give the empirical $\hbar$.

Quantitatively: if $\mathcal{B}_{\text{P04}}(s)$ is constant over a chain-step, then

$$\hbar = \mathcal{B}_{\text{P04}} \cdot \tau \implies \mathcal{B}_{\text{P04}} = \hbar/\tau \sim 10^{-34}/10^{-43} = 10^9 \text{ J/s} = 10^9 \text{ W},$$

i.e., gigawatt-level bandwidth density per substrate cell.

This is **structurally interpretable**: substrate cells operate at high bandwidth per cell but cells are extremely small in spacetime extent, giving a small per-step total. The macroscopic $\hbar$ is the time-integrated chain-step budget. *Note (per §3.3 circularity disclaimer): this numerical “check” is **tautological by construction** — it solves for $\mathcal{B}_{\text{P04}}$ given $\hbar$ rather than computing $\hbar$ from $\mathcal{B}_{\text{P04}}$; it is a relabeling, not a computation.*

3.7 Future substrate-derivation route

A first-principles substrate derivation of $\hbar$ would require:

- Substrate-derivation of $\mathcal{B}_{\text{P04}}(s)$ from substrate primitives (currently INHERITED).
- Substrate-derivation of chain-step duration τ (currently inherited via $c + \ell_P$).
- Composition to give $\hbar$ in absolute units.

This is an **open** derivation, analogous to Paper_ED_SC_4.2 Route A. Closure would upgrade the verdict for $\hbar$ from value-INHERITED to value-FORCED.

4 Audit Table

#	Step	Label	Notes
1	P01 chains	P	Primitive.
2	P04 bandwidth	P	Primitive.
3	P05 polarity-transport	P	Primitive.
4	P11 commitment-irreversibility	P	Primitive.
5	P03 + P13 substrate homogeneity	P	Primitives.
6	Action dimensions = bandwidth $\times$ time	I	Standard math.
7	Chain-step has substrate-time + polarity + bandwidth	D	Composition of primitives.
8	Participation budget per chain-step has dim. of action	D	Step 6 + step 7.
9	Chain-step invariance under P03 + P13	D	From P03 + P13.
10	$\hbar$ identification as participation quantum per step	D	Composition.
11	$\hbar$ structural form FORM-FORCED	D	From step 10.
12	$\hbar$ numerical value	I	Empirical matching.
13	$\mathcal{B}_{\text{P04}}(s)$ and τ INHERITED	I	Substrate parameters.
13a	Indep. substrate definition of $\mathcal{B}_{\text{P04}}$	OPEN	Relabeling, not computation.
14	Connection to standard QM	I	Standard QM.
15	Verdict M3	A $\rightarrow$ position	Per Paper_095.

5 Falsification Criteria

- **F1: Empirical detection of $\hbar$ variation across loci or epochs** (beyond cosmological-time variation of fundamental constants, which is itself a tested constraint). Would refute chain-step invariance.
- **F2: Substrate evidence that participation budget per chain-step does NOT have dimensions of action.** Would refute the dimensional structure.
- **F3: Empirical determination of substrate $\mathcal{B}_{P04}$ value inconsistent with the empirical $\hbar$.** Would force re-examination of the structural identification.
- **F4: $\hbar$ derived from first principles** (upgrade, not refutation) — would upgrade verdict to value-FORCED.

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

This paper supplies the substrate-level structural identification of $\hbar$ as participation quantum per chain-step. The structural form is FORCED; the numerical value is INHERITED.

Relationship to other ED papers:

- **Paper_002 (Born-Gleason):** $\hbar$ appears in probability normalization.
- **Paper_RQM_T8 (canonical commutation):** $\hbar$ is the rate constant in $[\hat{x}, \hat{p}] = i\hbar$.
- **Paper_012 (P-RB-1):** $\hbar$ in $E/\hbar$ atomic-superposition rate of becoming.

Numerical content INHERITED. $\hbar$ value, substrate $\mathcal{B}_{P04}$, chain-step duration. **Form FORCED.** $\hbar$ structural origin as participation quantum per chain-step.

Future work. Substrate-level derivation of $\hbar$'s value (highest-leverage open derivation in fundamental-constant sector; parallel to Paper_ED_SC_4.2 Route A in scope).

Verdict: **M3** (value-INHERITED with explicit structural-origin).